An Extension-Operator Positive Subcase for Complemented Copies in $C(\beta\omega \times \beta\omega)$

arXiv automatic math run `fa_banach_001`, lane 0

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Status

Partial result, likely valid. The packet records a positive subcase of Problem 4.1 of Plebanek, Rondos, and Sobota [1]. It does not solve the problem for arbitrary separable compact spaces.

Source Problem

The source paper is arXiv:2405.19120, *Complemented subspaces of Banach spaces $C(K \times L)$*. The problem appears on page 12 of the arXiv PDF.

> We start the final section of the paper by asking the following question, related to Corollary 1.4. Recall here that, for any separable compact space K, the space $\beta\omega \times \beta\omega$ maps continuously onto K and hence $\mathcal{C}(K)$ embeds isometrically into $\mathcal{C}(\beta\omega \times \beta\omega)$. We are however not aware of any example of a separable compact space K such that $\mathcal{C}(\beta\omega \times \beta\omega)$ does not contain a complemented copy of $\mathcal{C}(K)$.
>
> **Problem 4.1.** *Does $\mathcal{C}(\beta\omega \times \beta\omega)$ contain a complemented isomorphic copy of $\mathcal{C}(K)$ for every separable compact space K?*
>
> A natural and more general variant of Problem 4.1 could ask for a characterization of those separable compact spaces K for which $\mathcal{C}(K)$ is not isomorphic to a complemented subspace of $\mathcal{C}(\beta\omega \times \beta\omega)$.

Problem 4.1 asks:

$$\text{Does } C(\beta\omega \times \beta\omega) \text{ contain a complemented isomorphic copy of } C(K)$$

for every separable compact space K?

Definitions

Say that a compact space K is *cube-complemented at weight κ* if $C(K)$ is isomorphic to a complemented subspace of $C([0,1]^\kappa)$. Equivalently, there are bounded operators

$$J\colon C(K) \longrightarrow C([0,1]^\kappa), \qquad Q\colon C([0,1]^\kappa) \longrightarrow J[C(K)]$$

such that J is an isomorphic embedding and Q is a projection.

A concrete sufficient condition is the existence of a bounded linear extension operator. If K is a closed subspace of $[0,1]^\kappa$, $R\colon C([0,1]^\kappa) \to C(K)$ is the restriction map, and there is a bounded linear operator $E\colon C(K) \to C([0,1]^\kappa)$ with $RE = I_{C(K)}$, then $E[C(K)]$ is complemented in $C([0,1]^\kappa)$ by the projection ER. In particular this applies when K is a retract of $[0,1]^\kappa$, by taking $Ef = f \circ r$ for a retraction $r\colon [0,1]^\kappa \to K$.

Proof Intuition

Plebanek, Rondos, and Sobota prove that $C(\beta\omega \times \beta\omega)$ already contains complemented copies of all cube spaces $C([0,1]^\kappa)$ for $1 \le \kappa \le \mathfrak{c}$. Therefore any compact K whose $C(K)$-space is complemented inside one of those cube spaces can be inserted into $C(\beta\omega \times \beta\omega)$ by composing two complemented embeddings. The key point is simply transitivity of complemented subspaces, with the extension operator giving the first complemented embedding and the source paper giving the second.

Main Result

Theorem 1. *Let K be a compact Hausdorff space. Suppose that for some cardinal $1 \le \kappa \le \mathfrak{c}$, the Banach space $C(K)$ is isomorphic to a complemented subspace of $C([0,1]^\kappa)$. Then $C(\beta\omega \times \beta\omega)$ contains a complemented subspace isomorphic to $C(K)$.*

Proof. By Corollary 1.4 of [1], applied with $\Gamma = \omega$, there is an isomorphic embedding

$$A\colon C([0,1]^\kappa) \longrightarrow C(\beta\omega \times \beta\omega)$$

whose range is complemented. Let

$$P\colon C(\beta\omega \times \beta\omega) \longrightarrow A[C([0,1]^\kappa)]$$

be a bounded projection.

By hypothesis, choose an isomorphic embedding

$$J\colon C(K) \longrightarrow C([0,1]^\kappa)$$

and a bounded projection

$$Q\colon C([0,1]^\kappa) \longrightarrow J[C(K)].$$

Then $AJ\colon C(K) \to C(\beta\omega \times \beta\omega)$ is an isomorphic embedding. Define, for $F \in C(\beta\omega \times \beta\omega)$,

$$\Pi F = A\,Q\,A^{-1}(PF),$$

where A^{-1} is the bounded inverse from $A[C([0,1]^\kappa)]$ onto $C([0,1]^\kappa)$. This is a bounded linear operator on $C(\beta\omega \times \beta\omega)$. Its range is $AJ[C(K)]$, and for every $g \in J[C(K)]$,

$$\Pi(Ag) = AQg = Ag.$$

Thus Π is a projection of $C(\beta\omega \times \beta\omega)$ onto $AJ[C(K)]$. Hence $C(K)$ is isomorphic to a complemented subspace of $C(\beta\omega \times \beta\omega)$. □

Corollary 1. *Problem 4.1 has a positive answer for every separable compact K which admits a bounded linear extension operator from some cube $[0,1]^\kappa$ with $1 \le \kappa \le \mathfrak{c}$. In particular, it has a positive answer for every separable compact retract of such a cube.*

Proof. The extension-operator hypothesis makes $C(K)$ complemented in $C([0,1]^\kappa)$, as noted above, and the theorem applies. □

Verification Notes

The proof is purely operator-theoretic. The only imported mathematical input from the source paper is Corollary 1.4, which states that $C(\beta\omega \times \beta\omega)$ contains complemented copies of $C([0,1]^\kappa)$ for every $1 \le \kappa \le \mathfrak{c}$. The remaining step is the standard fact that the composition of complemented embeddings is again a complemented embedding, written explicitly in the proof by the projection $\Pi = AQA^{-1}P$.

The crop was generated by:

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conda run -no-capture-output -n sandbox python code/make_problem_crop.py.
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Limitations

This result does not address separable compact spaces K for which no complemented embedding of $C(K)$ into a cube space $C([0,1]^\kappa)$, $\kappa \le \mathfrak{c}$, is known. That remaining case is exactly where the source problem is hard: one needs either a new averaging/projection mechanism from $\beta\omega \times \beta\omega$ onto K, or an obstruction showing that no such mechanism can exist.

The source paper already proves the metric compact case, and more generally records complemented copies of $C(K^\kappa)$ for metric compact K and $\kappa \le \mathfrak{c}$. The present packet should therefore be read as a structural positive criterion and extension-operator/retract subcase, not as a full solution of Problem 4.1.

Novelty Check

Local lightweight indexes `registry_index.tsv`, `solutions/index.tsv`, `attempts/index.tsv`, and `proof_gaps/index.tsv` were searched for arXiv:2405.19120, the title, and the core terms $C(\beta\omega \times \beta\omega)$, complemented subspace, separable compact, extension operator, and Dugundji. No existing packet for this exact source problem was found.

An external web search on June 16, 2026, using exact phrases from Problem 4.1 and variants of "$C(\beta\omega \times \beta\omega)$ complemented $C(K)$ separable compact" found the source paper and unrelated nearby complementability papers, but no later paper explicitly answering Problem 4.1. This is a bounded check, not an exhaustive bibliographic review.

Human Review Recommendation

Reviewers should verify that the intended application class is stated only under an actual extension-operator or retract hypothesis. The core projection calculation is short and should be checked against the source Corollary 1.4.

References

[1] G. Plebanek, J. Rondos, and D. Sobota, *Complemented subspaces of Banach spaces $C(K \times L)$*, arXiv:2405.19120, 2024.

[2] A. Pelczynski, *Linear extensions, linear averagings, and their applications to linear topological classification of spaces of continuous functions*, Dissertationes Mathematicae 58 (1968), 92 pp.